TEXORA · LECTURE SHEET

Textile Composites

When the fabric becomes the skeleton of a solid: definitions, the reinforcement and matrix partners, manufacturing routes from hand lay-up to resin transfer moulding, applications, and the rule-of-mixtures calculation that predicts stiffness.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 5 · LEVEL 4 TERM 2

1. Definition

A **composite material** combines two or more distinct materials so that the combination has properties neither partner has alone. A **textile composite** (fibre-reinforced composite) uses a textile structure as the **reinforcement** — carrying load — inside a surrounding **matrix** — holding the fibres in place, transferring load between them, and protecting them from the environment. The definition to keep: *the fibres carry the load; the matrix keeps the fibres working together.*

- **Reinforcements:** glass, carbon, aramid and high-performance polymer fabrics — woven, multiaxial, braided, knitted and 3D structures (Topic 2), chosen by load directions.
- **Matrices:** thermosets (epoxy, polyester, vinyl ester — crosslinked, cannot be remelted) and thermoplastics (PP, PA, PEEK — meltable, tougher, recyclable in principle); also ceramic and metal matrices for extreme duties.
- **The interface** between fibre and matrix is the third partner: fibre surface treatments (sizings) decide whether load actually transfers or the fibres pull out like hair from clay.

2. Why composites win

The composite bargain is **specific properties**: stiffness and strength per unit weight. A glass/epoxy panel can match a steel panel's stiffness at a fraction of the mass; carbon/epoxy goes further, which is why aircraft, wind-turbine blades, boats, pressure vessels and sports goods are composite territory. Secondary wins: corrosion resistance (no rust), design freedom (curved, integrated shapes), fatigue behaviour, and tailored anisotropy — stiffness placed exactly along the load paths by orienting the textile layers, and nowhere else.

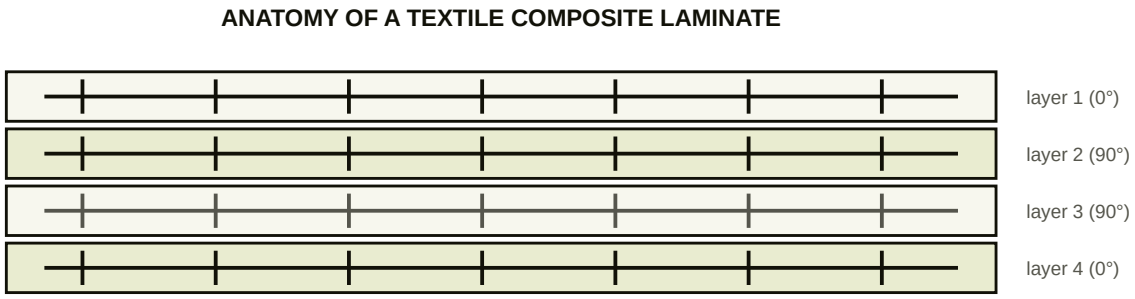


Figure 1 — Laminate construction: oriented textile plies embedded in matrix. Orientation is the composite designer's main variable.

3. Manufacturing processes

PROCESS	METHOD IN BRIEF	CHARACTER / TYPICAL PRODUCTS
Hand lay-up	Dry fabric placed in an open mould, resin applied by brush/roller, cured	Simplest, labour-heavy, variable quality; boats, tanks, prototypes
Spray-up	Chopped fibres and resin sprayed together into the mould	Fast, low fibre alignment; panels, baths
Vacuum bagging / infusion	Fabric stack sealed under a membrane; vacuum draws resin through	Better fibre fraction and consistency; wind blades, marine parts
Resin transfer moulding (RTM)	Dry preform in a closed mould; resin injected under pressure	Both faces finished, repeatable; automotive and aerospace parts
Prepreg / autoclave	Fabric pre-impregnated with resin, laid up, cured under heat and pressure	Highest performance and cost; aircraft structures
Filament winding	Resin-wet fibre wound onto a rotating mandrel in programmed paths	Pressure vessels, pipes, tanks — superb hoop strength
Pultrusion	Fibres pulled continuously through resin and a heated die	Constant-section profiles: rods, beams, gratings
Compression moulding	Sheet/bulk moulding compound pressed in matched dies	High-volume panels and housings

Table 1 — Composite manufacturing routes. Moving down the table generally trades labour for capital, and variability for repeatability.

Textile-side decisions inside every route: fabric areal weight and orientation per ply, fibre volume fraction (typically 40–60% — higher gives stiffness, but risks dry spots), void content (voids are strength thieves), and cure cycle. A composite is manufactured, not assembled: the material and the part are made in the same operation, which is why process control *is* quality control here.

4. Worked example — rule of mixtures

For a unidirectional layer loaded along the fibres, the composite modulus is estimated by the **rule of mixtures**:

$$E_c = V_f \cdot E_f + V_m \cdot E_m, \quad \text{with } V_f + V_m = 1$$

WORKED EXAMPLE — GLASS/EPOXY LAYER, $V_F = 0.50$

E-glass E_f = 72 GPa; epoxy E_m = 3 GPa.

$E_c = 0.50 \times 72 + 0.50 \times 3 = 36 + 1.5 =$ **37.5 GPa**.

Density by the same rule: $\rho_c = 0.50 \times 2{,}550 + 0.50 \times 1{,}200 =$ **1,875 kg/m³**.

Compare steel: E = 200 GPa at ρ = 7,850 kg/m³. Specific modulus: composite $37.5 \times 10^9/1{,}875 =$ **2.0×10^7** m²/s²; steel $200 \times 10^9/7{,}850 =$ **2.55×10^7** m²/s². In pure axial stiffness per weight, this glass laminate is already in steel's neighbourhood — and a woven fabric layer delivers roughly half the unidirectional value in each of two directions, which is exactly the trade Figure 1's stack makes.

5. Applications and honest limitations

Applications: aircraft and automotive structures, wind-turbine blades, boats and ships, pressure vessels and pipes, sporting goods, prosthetics and medical devices, civil repair and strengthening of concrete, ballistic panels (with aramid/HMPE). **Limitations to state without embarrassment:** low through-thickness strength (delamination), impact damage that hides inside the laminate, difficult repair and inspection, joining problems, higher material cost, and end-of-life — thermoset composites are hard to recycle, an active research problem (and one reason thermoplastic composites and recyclable resins attract attention).

Glossary

Reinforcement — the load-carrying textile phase. **Matrix** — the surrounding phase that holds and protects the reinforcement. **Prepreg** — fabric pre-impregnated with (partially cured) resin. **Preform** — a shaped dry textile ready for moulding. **Fibre volume fraction** — the share of the composite volume

occupied by fibre. **Delamination** — separation between plies, the characteristic composite failure. **Sizing** — the surface treatment on fibres that bonds them to the matrix.

Key points

• Fibres carry load; the matrix keeps them positioned, transfers load and protects them; the interface decides whether the partnership works.
• Composites are designed stacks: ply material, orientation and fibre fraction are the variables.
• Processes run from hand lay-up to autoclaved prepreg — capital buys repeatability and fibre fraction.
• Rule of mixtures gives first estimates of modulus and density; orientation roughly halves the unidirectional value per direction in a balanced woven layer.
• Delamination, hidden impact damage and thermoset recycling are the honest weaknesses.

Practice

1. A carbon/epoxy layer has V_f = 0.60, E_f = 230 GPa, E_m = 3 GPa, ρ_f = 1,800 kg/m³, ρ_m = 1,200 kg/m³. Calculate E_c, ρ_c and the specific modulus; compare with steel (2.55×10^7 m²/s²).
2. Choose a manufacturing process for (a) a production run of identical bus roof panels with both faces finished; (b) a one-off boat hull built in a small workshop. Justify each in one or two sentences.
3. Explain why void content and fibre volume fraction are process-control items, not just inspection items, in composites.

Answer key

1. E_c = 0.60 × 230 + 0.40 × 3 = 138 + 1.2 = **139.2 GPa**. ρ_c = 0.60 × 1,800 + 0.40 × 1,200 = 1,080 + 480 = **1,560 kg/m³**. Specific modulus = 139.2 × 10^9/1,560 ≈ **8.9×10^7 m²/s²** — about 3.5 times steel's value, the quantitative reason carbon composites dominate weight-critical structures.
2. (a) **RTM** (or compression moulding): closed mould gives both faces finished and repeatability across a production run. (b) **Hand lay-up**: minimal tooling cost and simple equipment suit a single large hull; the accepted cost is labour and property variability.
3. Because in composites the material is created during manufacture: once cured, voids cannot be removed and a low fibre fraction cannot be raised. Inspection can only reject the finished part; controlling resin flow, vacuum/pressure and cure during the process is the only way to *make* the specified material — quality is built in or not at all.